CIS 220 Database Concepts
Normalization Assignment

Create a document to hold your submission for this assignment. There will be multiple tasks in this assignment and all task solutions should be added to your document, clearly labeled. **Please follow the instructions carefully, especially where they specify the different formats that are expected.**

Effects of Business Rules on Design: Here are a couple of examples of how a small change in business rules can affect database design.

Soccer Matches: A The local parks and recreation department runs a soccer league and collects basic statistics at the end of every match. You need to design the data tables for this problem.

Location Date Played					Referee Name Phone Number, Address				
Team 1 Name Sponsor		Score			Team 2 Name Sponsor		Score		
Player Name	Phone	Age	Points	Penal.	Player Name	Phone	Age	Points	Penal.

After talking with people involved you have come up with a set of business rules.

- A match is between two Teams
- A Team can be matched against many other Teams
- There is a single Ref per match
- A Ref can officiate many matches
- A player plays on a single Team
- A Team has many Players

And the following 3NF tables

Match(**MatchID (PK)**, DatePlayed, Location, *RefID(FK)*)
Score(**MatchID (PK), *TeamID(PK,FK)***, Score)
Referee(**RefID (PK),** Phone, Address, City, State, Zip)
Team(**TeamID (PK),** Name, Sponsor)
Player(**PlayerID (PK),** Name, Phone, DoB, *TeamID(FK)*)
PlayerStats(***MatchID(PK,FK), PlayerID(PK,FK)***, Goals, Penalties)

Modeling the following functional dependencies

F = {

MatchID -> DatePlayed, Location, RefID

MatchID + TeamID -> Score

RefID -> Phone, Address, City, State, Zip

TeamID -> Name, Sponsor

PlayerID -> Name, Phone, DoB, TeamID

MatchID + PlayerID -> Goals, Penalties

}

After taking your initial ideas back for confirmation to the folks at the recreation department, you all realized that there was a slight problem with your interpretation. They acknowledge that the form they gave you is not the perfect description. You all agreed on the following change to the business rules.

- There can be multiple refs for a single match
- A Player can play on many teams

Sometimes if a team does not have enough players, one or two will switch to fill the roster out. There have even been cases where a player would play part of the game on one team and then have to switch due to injuries. They want to keep track of these details

TASK ONE: Your task is to modify the original (provided) table structures to handle these changes. Show

- the new business rules
- the new set of functional dependencies
- the new table listing using the same format demonstrated above.

Veterinary Visits: A local vet has asked you to design a database to track their animals, visits and the procedures. They have a few vets who are employed here.

After talking with people involved you have come up with a set of business rules.

- An Animal is assigned a single Category
- A Category can have many Animals
- An Animal may schedule many Visits
- A Visit is for a single Animal
- A Visit may have one Procedure
- A Procedure can be performed on many Visits
- A Vet can attend many Visits
- Each Visit is attended by a single Vet

Category(**CategoryID(PK),** Name, Descrption)
Animal(**AnimalID(PK),** Name, *CategoryID(FK)*, Breed, DateBorn, Gender, Registered, Color, Photo, PetStore)
AnimalVisit(**VisitID(PK),** *AnimalID(FK), VeterinaryID(FK), ProcedureID(FK),* Cost, VisitDate)
Veterinarian(**VeterinaryID(PK)**, First_name, Last_name, Address, ZipCode, CityID, Phone)
Procedure(**ProcedureID(PK),** Name, Description)

Modeling the following functional dependencies

F = {
CategoryID -> Name, Description
AnimalID -> Name, Category, Breed, DoB, Gender, Registered, Color, Photo, PetStore
VisitID -> AnimalID, VetID, ProcedureID, VisitDate, Cost
VeterinaryID -> First_Name, Last_Name, Address, ZipCode, CityID, Phone
ProcedureID -> Name, Description
}

After taking your initial ideas back for confirmation to the folks at the recreation department, you all realized that there was a slight problem with your interpretation. You all agreed on the following changes.

- They want to be able to record all the breeds for mixed-breed animals
- A visit can have multiple procedures performed
- They also want to begin tracking genealogical records of their animals by noting the mother and father.

TASK TWO: Modify the original table structures to handle these changes. Show the new business rules, new set of functional dependencies and the new table listings.

TASK THREE: Reason about and normalize a poorly designed table

staffNo	dentistName	patientNo	patientName	appointment		surgeryNo
				date	time	
S1011	Tony Smith	P100	Gillian White	12-Aug-03	10.00	S10
S1011	Tony Smith	P105	Jill Bell	13-Aug-03	12.00	S15
S1024	Helen Pearson	P108	Ian MacKay	12-Sept-03	10.00	S10
S1024	Helen Pearson	P108	Ian MacKay	14-Sept-03	10.00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Oct-03	16.30	S15
S1032	Robin Plevin	P110	John Walker	15-Oct-03	18.00	S13

Figure 1: Details of patient dental appointments.

The table shown in Figure 1 is susceptible to modification anomalies. **Provide examples of insertion, deletion, and update anomalies.**

Describe and illustrate the process of normalizing the table to 3NF. State any assumptions you make about the data shown in this table. I want to see you take the table through appropriate structural changes to 1NF, then 2NF, then 3NF.

Start by listing the functional dependencies and then resolve by modifying the table structure.

Show the table structures in the following format

Table_Name(**primary_key**, attribute1, attribute2, attributeN, *foreign_key)*

Legend

- Bold and underline -> primary key
- Italicized -> foreign key

TASK FOUR: Determine the status of a table and normalize if necessary.

Consider the relation schema ***Membership*** for a library database as follows:

Membership (**MID**, Name, Address, PhoneNum, Parent, ISBN, Title, Authors, BorrowDate, ReturnedDate, FineDue, FinePaid).

Note: Here, Parent may have the values Null, Father_Name, Mother_Name or both.

State the normalization rule violations if any.

- Is the table in 1NF, why or why not?
- Is the table in 2NF, why or why not?
- Is the table in 3NF, why or why not?

If there are any violations, normalize the Membership schema to 3NF and **show the steps**. Start by listing the functional dependencies and then resolve by modifying the table structure.

Show the table structures in the following format

Table_Name(**primary_key**, attribute1, attribute2, attributeN, *foreign_key)*

Legend

- Bold and underline -> primary key
- Italicized -> foreign key